

P-hacking with Random Data Teacher Notes

TOK Cognitive Science Activity Bank • Natural Sciences • 35-45 min

Category	Natural Sciences	Time	35-45 min
Difficulty	Extended	ID	p-hacking-with-random-data

Big idea: Methods can create false confidence.

Overview

Students experience how false patterns can appear when many tests are tried.

Student Prompt

Inspect P-hacking with Random Data Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

P-hacking with Random Data Stimulus A	Student-facing stimulus 1: A random dataset where students hunt for any impressive correlation. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
P-hacking with Random Data Stimulus B	Student-facing stimulus 2: Multiple subgroup comparisons that eventually produce a 'significant' pattern. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
P-hacking with Random Data Reveal / Twist	Student-facing stimulus 3: A publication pitch followed by a methods audit. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Methods can create false confidence.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting P-hacking with Random Data Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from P-hacking with Random Data Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Can correct methods lead to false knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how methods can create false confidence.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: A random dataset where students hunt for any impressive correlation.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- A random dataset where students hunt for any impressive correlation.
- Multiple subgroup comparisons that eventually produce a 'significant' pattern.
- A publication pitch followed by a methods audit.

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups a random data table.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Let them test as many relationships as possible.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Reward groups for finding an apparently interesting pattern.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Reveal that, with enough searches, chance findings become likely.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may think observation happens before theory in a simple linear order.
- Students may overclaim from a classroom model without naming its limits.

Assessment Look-Fors

- Names variables, assumptions, controls, anomalies, and limits.
- Explains how methods shape what can count as scientific knowledge.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Observation is Theory-Laden
- Placebo Trial Design
- Model Failure Challenge